



Tilt Indicator

Abstract

In this assignment, you will design and implement a simple tilt indicator using the IMU sensor, the 595 shift register, and a set of eight LEDs. The goal of this project is to visualize the orientation of your breadboard by mapping its tilt angle to a corresponding LED pattern. This activity will help you practice integrating multiple hardware components, including sensors and digital output devices, while applying basic programming techniques to translate sensor readings into visual feedback. By completing this assignment, you will gain a deeper understanding of sensor-based motion detection, data mapping, and digital signal control.

Description

Your task is to build a system that uses eight LEDs to represent the tilt angle of your breadboard within a range of approximately plus or minus ninety degrees. When the board is tilted in either direction, the LEDs should illuminate in a way that clearly communicates the degree of tilt. The specific visual effect is flexible, and you are encouraged to be creative in deciding how the LEDs respond to changes in angle.

For instance, you might choose a cumulative lighting effect in which each LED turns on sequentially and remains lit as the tilt angle increases. In this case, when the board reaches one extreme of the range, all eight LEDs would be illuminated, while at the opposite extreme they would all be off. Alternatively, you may choose a moving light effect in which only one LED is lit at a time to represent the current tilt angle. The important goal is that the LEDs provide a clear and consistent visual representation of the board's orientation within the full tilt range.

To demonstrate the functionality, you should test the system by tilting your breadboard gradually from one side to the other, covering the full range of approximately plus or minus ninety degrees. Observe how the LEDs respond and verify that the lighting pattern accurately reflects the direction and magnitude of the tilt.

Because the available breadboard space is limited, you will need to connect the MPU9250 to the system using jumper wires. The IMU can be secured to the breadboard using a simple method such as blue tape or by creating a 3D-printed mount. The key requirement is that the IMU moves together with the breadboard so that its readings accurately correspond to the physical tilt.

Requirements

1. The LEDs only need to respond to tilting along one axis, for example, the X-axis. You do not need to measure tilt in multiple directions for this assignment.
2. You may use Arduino's built-in functions such as `shiftOut()`, `bitClear()`, and `bitSet()` to control the shift register and manage the LED output efficiently.

Delivery

There are two deliverables.

- First, your code. You can put them in a zip file if needed.
- Second, a quick video demo of your experiment.

Due Date

Thursday April 30th, 11:59 PM EST